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7.1: Using the gm values from assignment 5.

Parasitics around C_c are comparable to C_c itself (1pF) and nearly double its effective value, due to which $\omega_{u,loop}$ becomes nearly half.

For example, C_{gd} of M_{11} , which directly comes across C_c is around 529.7 fF.

This can be compensated for by increasing the gms or by decreasing C_c to bring $\omega_{u,loop}$ back to the expected value.

Procedure:-

- Slew rate was found using the slewRate function of the calculator in cadence.
- The swing limits were taken as the input DC voltages for which the output was found to not remain linear and saturate to the supply rails.
- HD_3 was found from the ratio of the amplitudes of the 3rd harmonic and the fundamental component from the dft option of the calculator in cadence.

Pictures of plots and testbenches/schematics used have been attached at the end.

a)

Specifications and component value

Specification	Value	Unit
Closed loop dc gain k	8	-
Closed loop bandwidth f_B	20	MHz
Load capacitor C_L	24	pF
Load resistor R_L	663.14	Ω
Input resistance R_i	10	k Ω
C_c	1	pF
Component parameter	Value	Unit
L (all)	0.3	μm
W_{00}	1	μm
m_{00}	2	-
I_{00}	79	μA
$V_{dsat_{00}}$	251.6	mV
W_0	0.5	μm
m_0	11	-
V_{dsat_0}	252.9	mV
$W_1=W_2=W_5=W_6$	1	μm
$m_1=m_2=m_5=m_6$	10	-
V_{dsat_1}	151.6	mV
V_{dsat_2}	151.8	mV
V_{dsat_5}	146	mV
V_{dsat_6}	146.8	mV
VB56	1.13	V
V_{DS_1}	200.4	mV
V_{DS_2}	198.5	mV
$W_3=W_4=W_7=W_8$	1	μm
$m_3=m_4=m_7=m_8$	45	-

Vdsat ₃	-147.3	mV
Vdsat ₄	-147.3	mV
Vdsat ₇	-147.5	mV
Vdsat ₈	-146.2	mV
VB78	0.95	V
VDS ₃	-199.4	mV
VDS ₄	-201.2	mV
W ₁₂	1	μm
m ₁₂	166	-
Vdsat ₁₂	252	mV
W ₁₁	1	μm
m ₁₁	850	-
Vdsat ₁₁	-249.9	mV

Transconductances

Transconductance	Value	Unit
Gm1 (assignment 5)	1.13	mS
gm ₁	0.97	mS
gm ₂	0.97	mS
Gm ₂ (assignment 5)	40.87	mS
gm ₁₁	41.5	mS

b)

Result	Value
Closed loop DC gain	18.05 dB = 7.989
Closed loop 3-dB bandwidth	10.3726 MHz
Unity loop gain frequency	8.9414 MHz
Phase margin	83.316°
rms output noise	0.67 mV

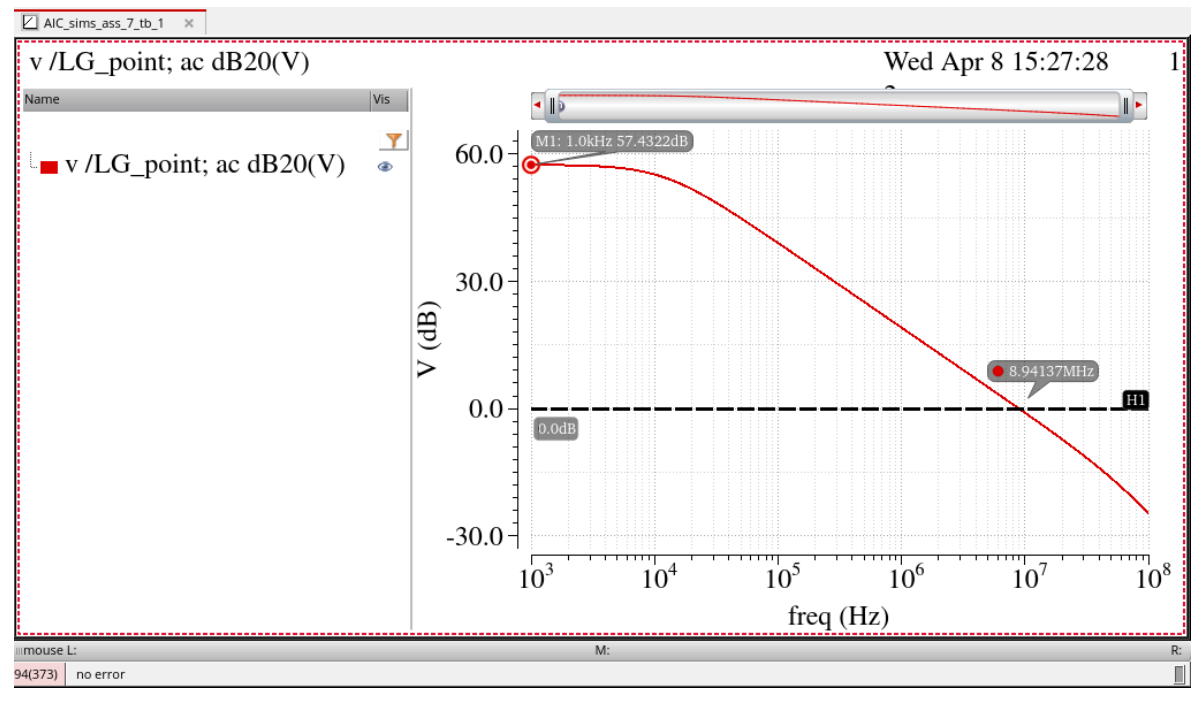
Opamp open loop DC gain	57.462 dB = 746.62
Positive slew rate	125.3 MV/s
Negative slew rate	94.99 MV/s
Positive swing limit	100 mV
Negative swing limit	-100 mV
HD ₃	0.0436 = -27.21 dB
Supply voltage	1.8 V
Current consumption	6.885 mA

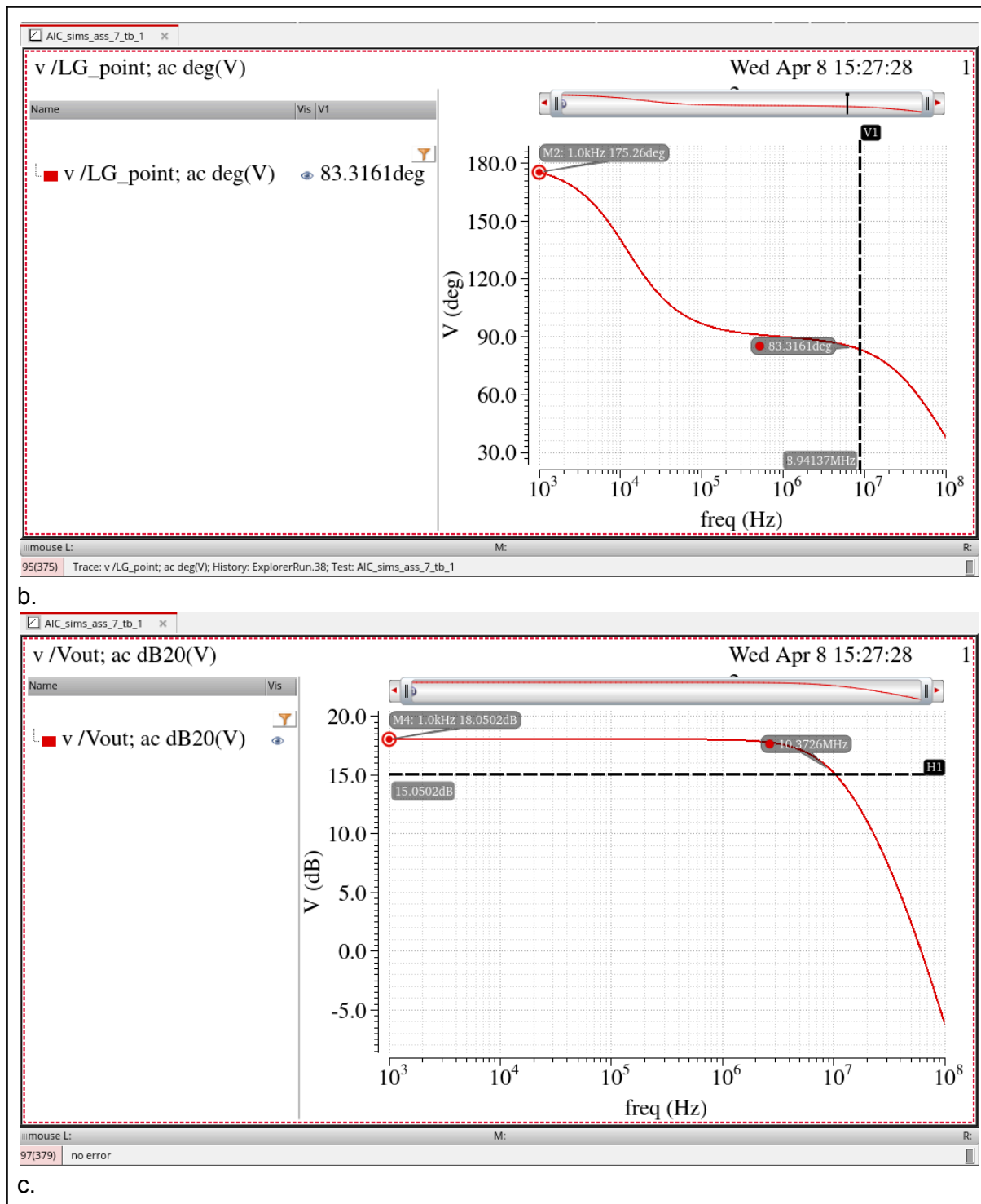
Noise variance contributions

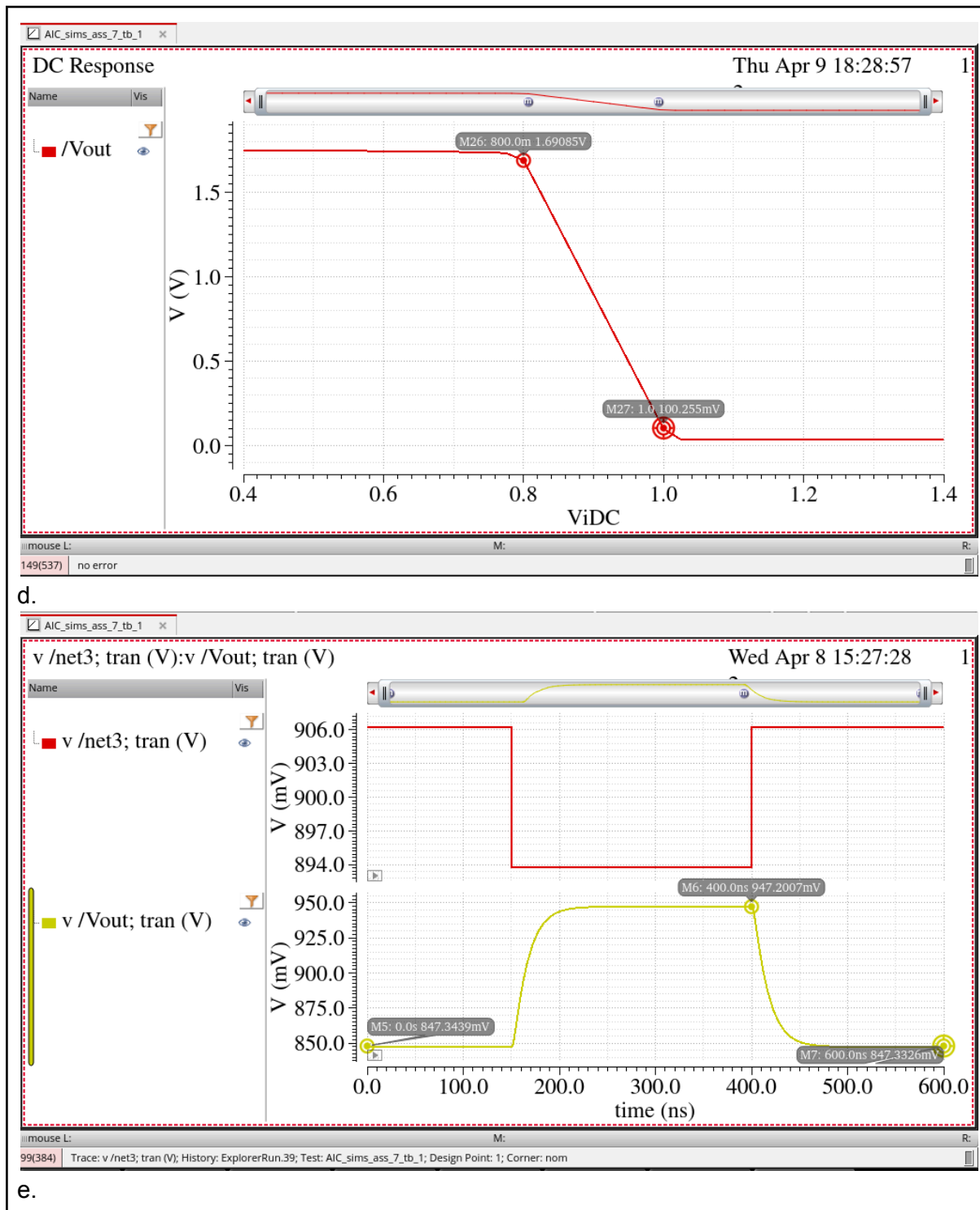
Component	Percentage
Ri	41.52
Rf	5.19
First stage (M ₁ -M ₈)	50.56
Second stage (M ₁₁)	0.02
RL	0.00

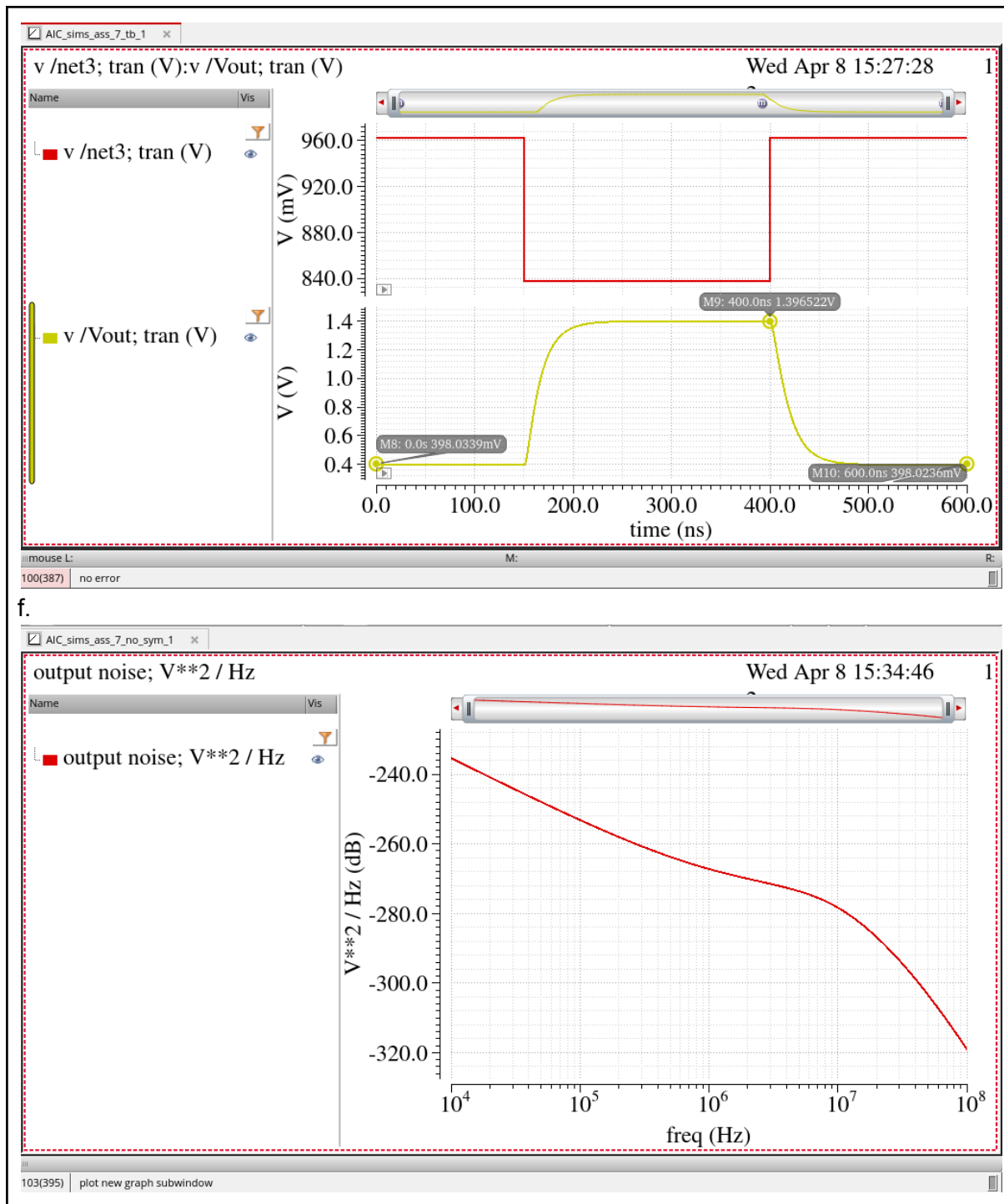
c) Plots:-

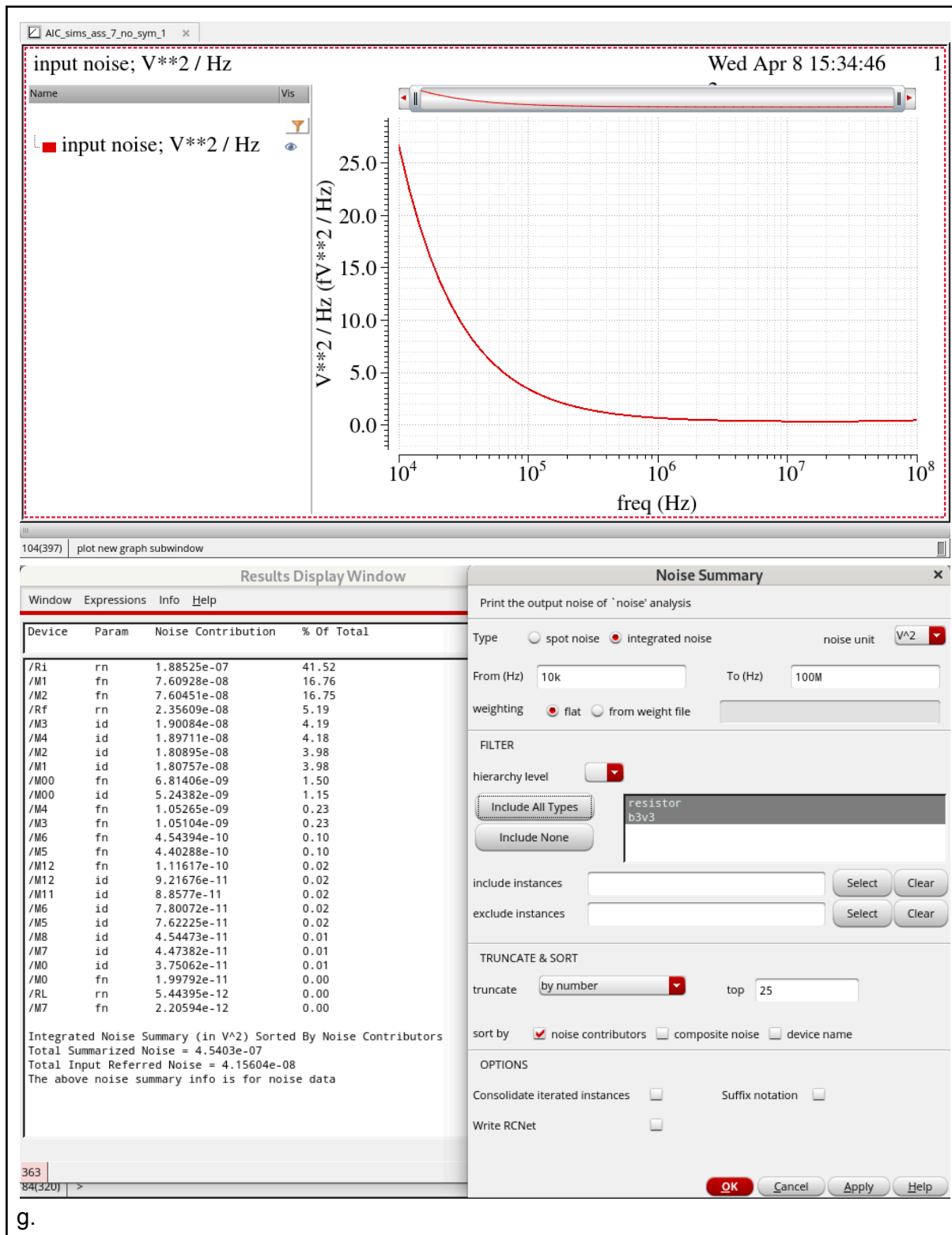
a.



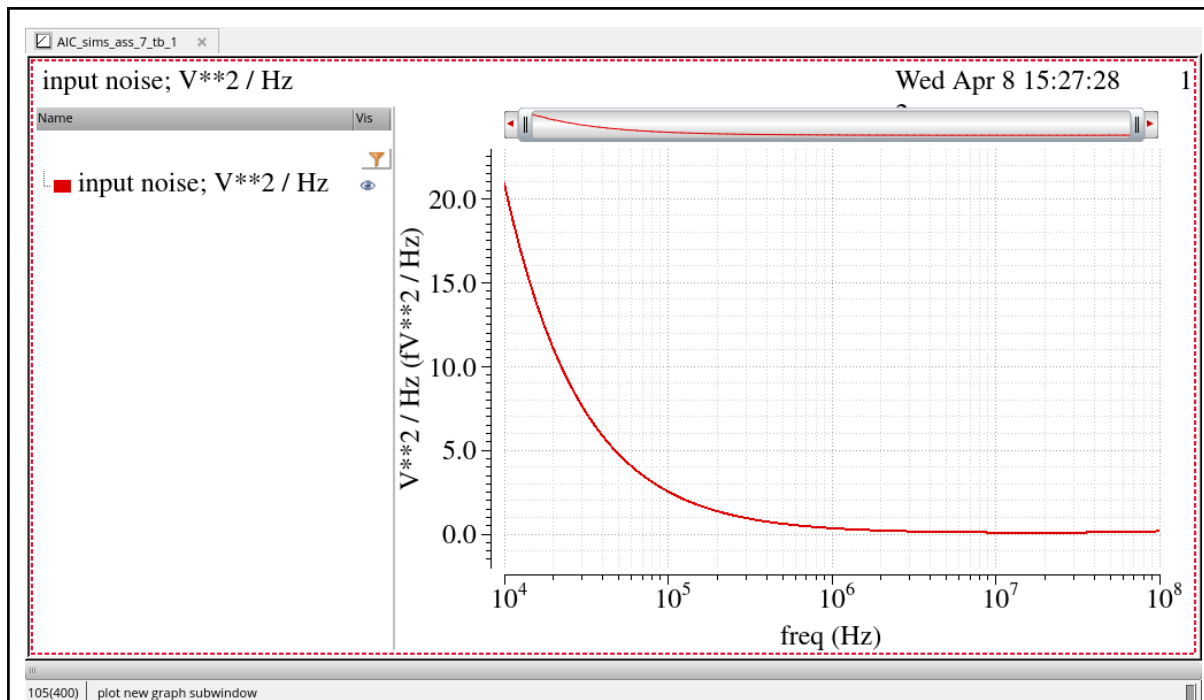






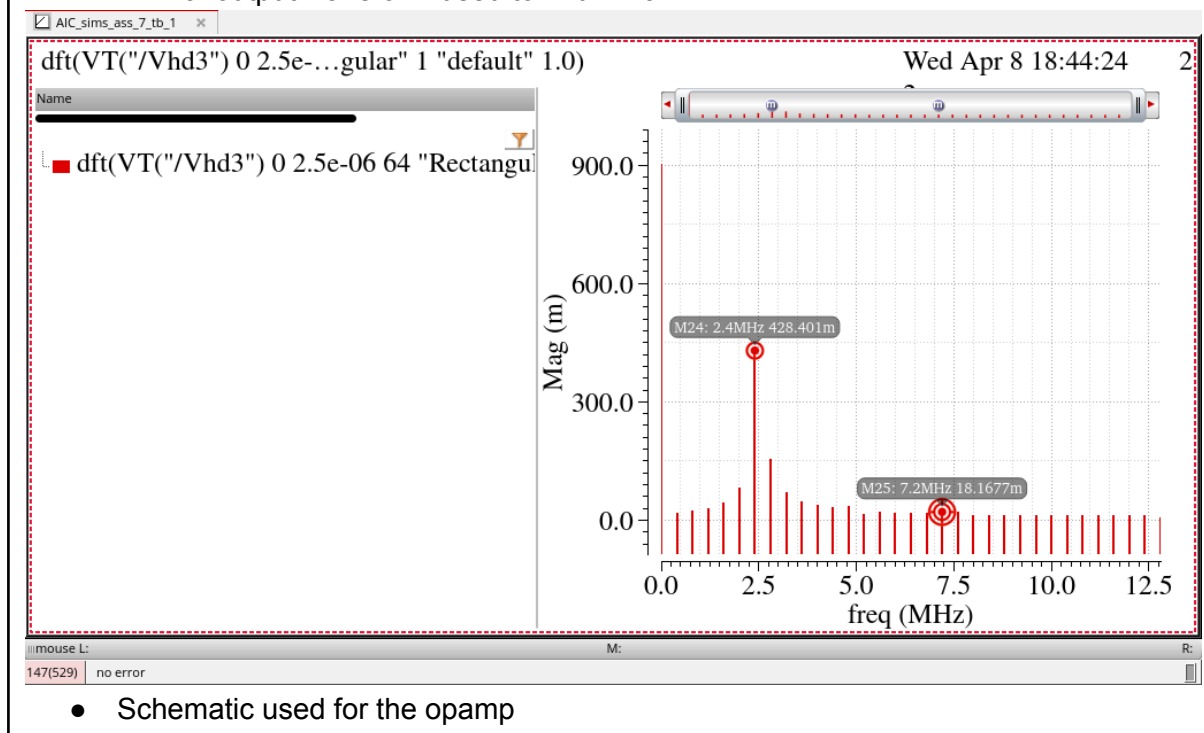


g.

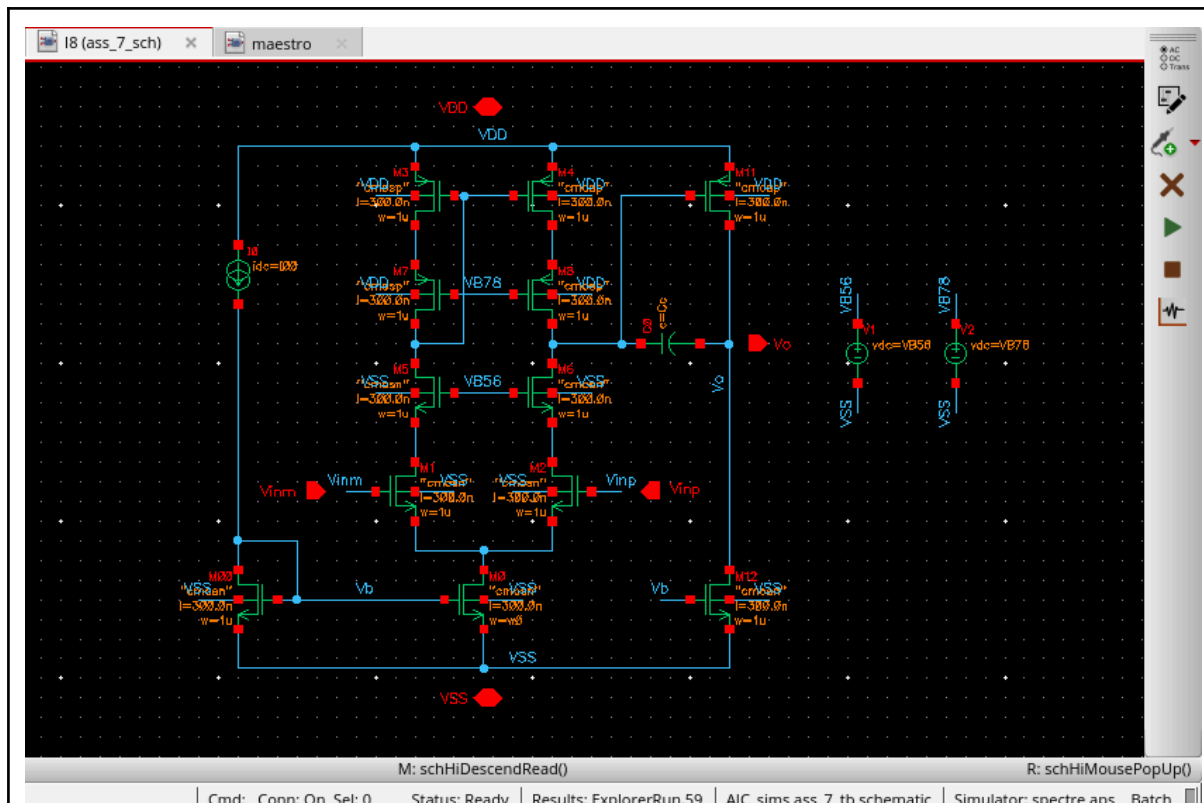


Miscellaneous

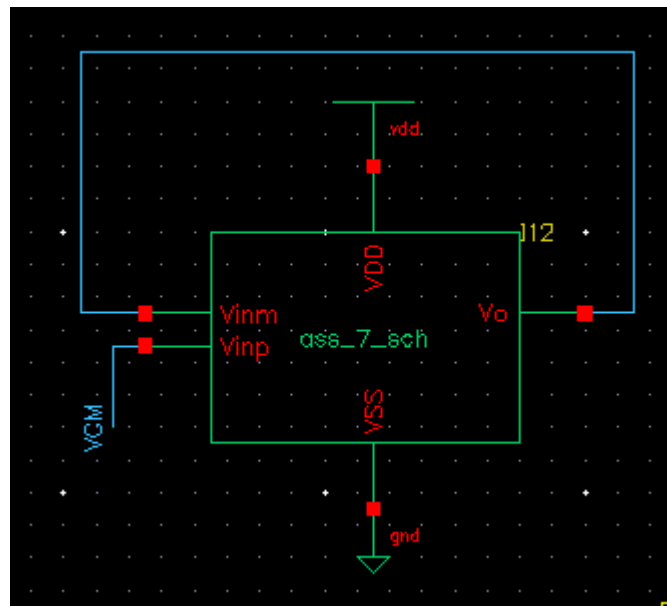
- DFT of output waveform used to find HD3:-



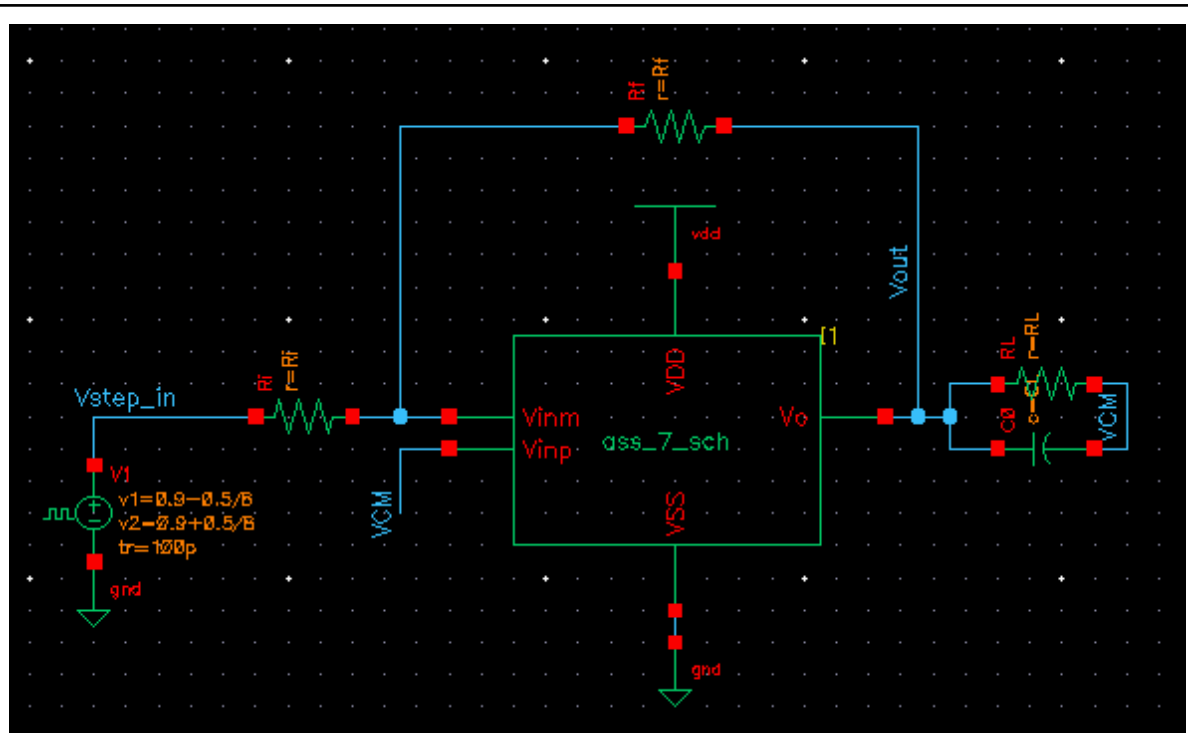
- Schematic used for the opamp



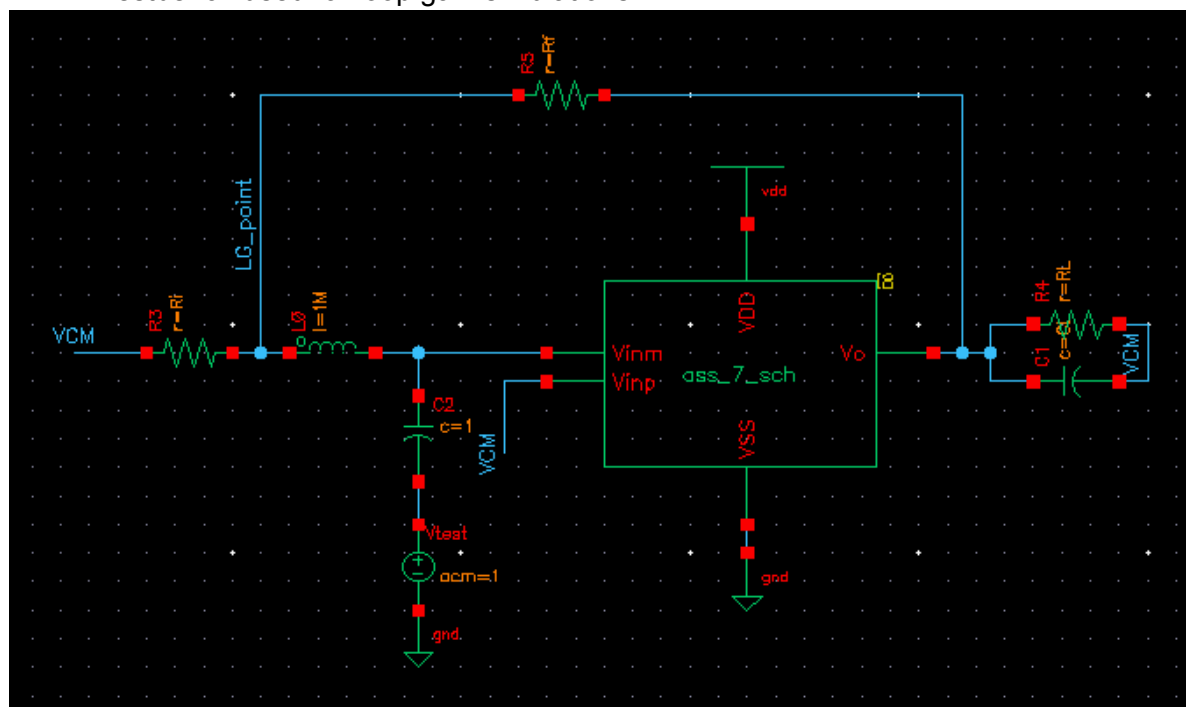
- Testbench used for DC simulations



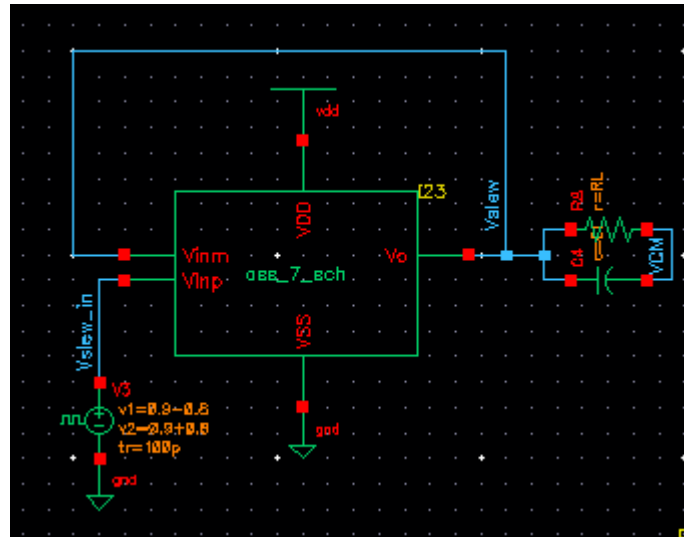
- Testbench used for CLTF/step response simulations



- Testbench used for loop gain simulations



- Testbench used for slew rate simulations



- Testbench used for HD3 simulations

